

Samuel King

PHD STUDENT · GEORGETOWN UNIVERSITY

✉ sik29@georgetown.edu | 🏠 samuel-king.org | 🔗 linkedin.com/in/sam-king-phd/

Professional Summary

Fifth-year PhD candidate in theoretical computer science with research experience in algorithms and time-space tradeoffs in data structures. Skilled in **abstract reasoning, edge case analysis, and modeling/prediction**. Six years of AI experience through rigorous coursework, personal projects, and engagement with academic literature. Currently transitioning to research in **mitigating AI risk**.

Education

Georgetown University

PHD, COMPUTER SCIENCE

- Advisor: Sasha Golovnev
- Research Interests: Algorithms, Data Structures, Discrete Math, Complexity

Washington D.C.

Fall 2021 - present

University of Rochester

HONORS BS, MATHEMATICS

- Computer Science Minor
- Thesis: Sampling Young Tableaux and Contingency Tables
- Take Five Scholar: The Role of Music in Video Games and Film
- Magna Cum Laude, GPA: 3.96

Rochester, NY

Fall 2016 - Spring 2021

Work Experience

Predict Health, Inc.

DATA SCIENTIST

- Researched and developed retention models for Medicare populations to expand Predict Health's data science capabilities
- Built a custom Python package to ensure efficient, reusable capabilities for future retention projects
- Streamlined codebase to reduce project completion time from months to days
- Designed experiments to detect and mitigate data leakage in temporal prediction tasks

Arlington, VA

May 2025 - Present

Publications & Patents

Authors listed in alphabetical order

Karthik Gajulapalli, Alexander Golovnev, Samuel King, and Sidhant Saraogi. Online Orthogonal Vectors Revisited. To appear in *Symposium on Discrete Algorithms (SODA)*, 2026.

Karthik Gajulapalli, Alexander Golovnev, and Samuel King. On the Power of Adaptivity for Function Inversion. *Conference on Information-Theoretic Cryptography (ITC)*, 2024.

Aniruddha Bapat, Andrew M. Childs, Alexey V. Gorshkov, Samuel King, Eddie Schoute, and Hrishee Shastri. Quantum routing with fast reversals. *Quantum*, 5, 533, 2021.

U.S. Patent Application 18178491, filed Mar 3, 2023.

U.S. Patent Application 18669111, filed May 20, 2023.

Erin Bevilacqua, Samuel King, Jürgen Krietschgau, Michael Tait, Suzannah Tebon, and Michael Young. Rainbow numbers for $x_1 + x_2 = kx_3$ in $\mathbb{Z}_n$. *Integers: Electronic Journal of Combinatorial Number Theory*, 20(A50):A50, 2020.

Selected Coursework & Skills

Mathematics: Combinatorial Optimization, Spectral Graph Theory, Linear Algebra, Computational Statistics, Combinatorics, Algebra I & II

Computer Science: Algorithms, Data Structures, Sampling Algorithms, Analytical Methods, Cryptography, Parallel Algorithms, Databases, Gems of Theoretical CS

Artificial Intelligence: Experimental AI, Neural Networks, Artificial Intelligence, Philosophy of AI, AI Safety Fundamentals

Skills: Python (primary language), R, Mathematica, SQL (familiar languages), Git, $\text{\LaTeX}$, PyTorch, NetworkX, Pandas, VSCode, Cursor, Databricks, Technical Communication

Projects

Reinforcement Learning for Ramsey Numbers

Georgetown University

PERSONAL RESEARCH PROJECT

Summer 2024

- Conducted research on RL approaches for finding counterexample graphs for lower bounds on Ramsey numbers
- Experimented with deep Q-learning, cross-entropy method, game-playing, and Monte Carlo tree search
- Experiments conducted using PyTorch, Gymnasium, Stable-Baselines3

Exploring Subitizing in Convolutional Neural Networks

Georgetown University

NEURAL NETWORKS CLASS PROJECT

Fall 2022

- Trained convolutional neural networks to count the number of simple shapes in an image
- Compared performance between classification and regression models
- Tested model robustness with different synthetic datasets

Stratego AI

University of Rochester

HONORS SEMINAR RESEARCH PROJECT

Spring 2018

- Implemented traditional state-space tree search algorithms to play the board game Stratego
- Experimented with using neural networks in Tensorflow to learn the state value function
- Experiments were unsuccessful, but the problem was solved by a team of 34 DeepMind scientists 4.5 years later

Teaching Experience

Gems of Theoretical CS, Teaching Assistant & Lecturer

Georgetown U, Fall '25

CS Department Tutoring, Head Tutor & Program Founder

Georgetown U, Spring & Fall '25

Data Structures, Guest Lecturer

Amherst College, Fall '24

Introduction to Algorithms, Teaching Assistant & Lecturer

Georgetown U, Fall '24

Math Methods for CS, Teaching Assistant & Lecturer

Georgetown U, Spring '24

Intro to Databases, Teaching Assistant & Lecturer

Georgetown U, Fall '23

New Horizons in Theoretical Computer Science, Teaching Assistant

TTIC (Virtual), Summer '23

Graduate Algebra I, Teaching Assistant & Lecturer

U of Rochester, Fall '20

Honors Calculus I & II, Teaching Assistant & Workshop Leader

U of Rochester, Fall '17-Spring '20

Honors and Awards

2025 **Outstanding Teaching Assistant Award**, Georgetown U. Computer Science Department

2024 **Outstanding Teaching Assistant Award**, Georgetown U. Computer Science Department

2020 **Arthur S. Gale Memorial Prize**, University of Rochester Mathematics Department

2019 **Take Five Scholarship**, University of Rochester

Deans' Award for Symposium Presentation, University of Rochester

Joint Mathematics Meetings 2019 Outstanding Poster Distinction, JMM 2019